

Building Classification Using a YOLO Object Detection Model Trained on Aerial Imagery

Gregory Macfarlane, Ph.D.*, Connor Williams*, Kaylen Turley*

* Department of Civil and Construction Engineering

Brigham Young University, Provo, Utah 84604

gregmacfarlane@byu.edu, willicon@byu.edu, kaylentu@byu.edu

Abstract—Travel forecasting models rely heavily on land use data from government agencies, which can be inconsistent or unavailable in many regions. In this paper we investigate the viability of using aerial imagery and machine learning models as a first step to develop a method to obtain this statistical data.

Using Logan, Utah as a primary case study, we used the Microsoft Building Footprints dataset and aerial imagery to train a “You Only Look Once” (YOLO) model to classify building types, including residential, commercial, and apartment buildings. We found that the model demonstrated high accuracy in identifying building types in the training city. However, when validated against five other Utah cities selected for diversity of median income and housing ages, the model showed limited transferability, which suggests that differences between house architecture could necessitate local training.

We analyzed these performance discrepancies and discuss future potential research such as integrating additional environmental indicators and using high-resolution drone imagery to create more robust models for estimating household statistics and land use data.

Index Terms—aerial imagery; deep learning; AI; transportation; land use data

I. INTRODUCTION

Travel forecasting models rely on land use data as a primary input; these data include the number of persons and households living in a given area, as well as other information relevant to travel behavior including the availability of motor vehicles, the income of the residents, and the presence of workers, students, and non-workers. In many contexts, these data are assembled from government statistical agencies; in the United States common sources include the US Census Bureau and the Bureau of Labor Statistics. In countries where statistical agencies are not reliable, travel forecasting models are generally not developed or have data gaps. In countries where the data are available, these models are highly reliant on the consistent and public release of the data.

Recent research has sought to find alternative methods to collect land use characteristics. Satellite aerial imagery has been used to analyze the volume and density man-made structures and roads to estimate area populations, in both urban and rural settings [Allaw et al. [1]][2]. Day time imagery has also been used to track vehicle movement between major cities as an indication of economic activity [3]. Such imagery has also been used to classify land use by estimating landscape patterns and building functions, identifying different surfaces, and then

categorizing building types [4]. Night time light data has also been shown to be useful as well. One study correlated city light output with social economic variables such as population, GDP and eclectic power consumption [5], while another used the light emitted from cars at night on highways to predict vehicle travel between cities [6]. Satellite imagery has also been used to produce data sets such as the Microsoft Building Footprints data set, which has been used along with LiDAR and point of interest data to inventory residential structures from census data [Microsoft [7]][8]. Unfortunately, processes and methods reliant on satellite data are also susceptible to the availability of said data, which like statistical data may not be available across the world or could have its public access revoked.

With the rise of commercially available and affordable aerial drones, the ability to capture remote sensing data such as high-quality aerial imagery has become significantly easier. Over the past several decades this availability of data has led to high-resolution and accurate products that has benefited many fields [9]. One such field is the scientific sector, where drones have seen many applications [10]. Another advantage of aerial drones vis-à-vis satellite imagery is the ability to capture images from multiple angles, which can be used to create a more complete picture of a given area [11]. Both of these applications have been aided by the creation of open-source software for planning drone flights and analysing the collected data [12]. This wide adoption of drones is not without its own problems. While not being as reliant on the public release of data is satellite data, concerns over the wide spread use of drones in data collection is a debated topic. One of the biggest of these are the ethical concerns of collecting higher resolution data and its interaction with privacy laws [13].

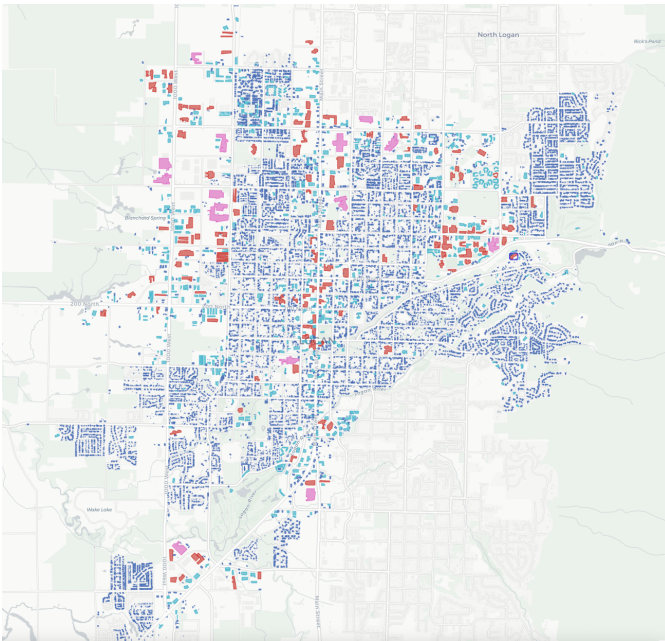
More recent advances in machine learning have led to more applications of remote sensing data, including tracking vehicle movement between cities [3], or predicting population characterizes and locations of man-made structures across entire countries [2]. Models such as Ultralytics’ “You Only Look Once” (YOLO) model [14] are being used to identify buildings and extract building footprints [Nurkarim and Wijayanto [15]][16], a particularly useful application for travel forecasting model inputs. It is expected that as travel forecast and other data sets continue to grow larger the use of machine learning models for analysis will become more and more relevant and practical.

Within the context of this background and research the question arises as to whether and to what extent remote sensing data can be used to develop land use independent from statistical agencies. The development of a process or method that could synthesize such data would solve many problems in places without reliable statistical data as well as supplement data in other places.

In this paper we present initial efforts to determine building type from building footprint data and aerial images of buildings. We do this by training a deep learning model on labeled data from Logan, Utah. We then apply this model to other cities in Utah to assess its viability in predicting building categories outside of the training data. The paper closes by discussing limitations and associated future research.

Methodology

A. Data Collection



We chose the city of Logan, Utah for the initial analysis due to it being smaller and having a variety of building ages and types.

With the chosen city the latitude and longitude of the official city boundaries was obtained. Using this boundary, all the building footprints in the city were extracted from the Microsoft Building Footprint dataset and saved.

This dataset allowed for easy access to building footprint data for this initial study.

The buildings were then classified using a simple K-mean analysis to differentiate different sized buildings.

To do this, the optimal K-mean was found using the elbow method. Using this K value, a K-mean analysis was run comparing the areas of the building footprints to their perimeters. **@fig-logan-clusters-map-plot** shows the distribution of different cluster types, with most of the buildings in Logan being a smaller size.

Using the building footprint location and the python Library “Contextily”[17], 600 randomly chosen aerial images of



Figure 1: YOLO prediction example

building in Logan were taken from publicly available satellite imagery and cropped to focus on a single building. 200 of the images were from clusters that included only larger buildings to help vary building types in the dataset.

B. Model Training

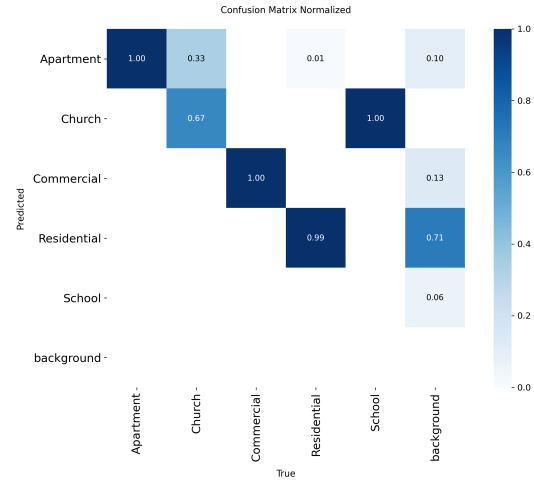
Ultralytics’ “You Only Look Once” or YOLO model is an open-source object detection model that has become very popular within the research community due it’s accuracy and ease of use. [14] The model is trained on images that have been manually annotated with bounding boxes, to identify the categories desired for the model to detect. The model trains itself on these images over hundreds of iterations until it can consistently recognize the different categories of objects. The model validates itself on other specified annotated images to ensure accuracy. The YOLO model is then able to predict these categories in nonannotated images by bounding them in bounding boxes and displaying the probability of correct identification. An example of a YOLO model prediction in Logan, Utah is shown in Figure 1.

Using the open-source tool Label Studio [18], the collected street map images were manually annotated. This annotation including creating bounding boxes around the roof of each building to categorize the building. The main buildings in each picture were labeled into five building categories: Apartment, Residential, Commercial, Church, and School. This order was constant across annotations. Best judgement was used in the annotation of the buildings, based off the appearance of the building and surrounding buildings and area. These bounding boxes were saved as text files and each one matched up with its corresponding image.

The annotated data sets were split into two categories. 80% of the images were used in training the object detection model, while 20% of the images were reserved for validation purposes. Within these image sets clusters that included larger and less common building sizes were specifically included, due



(a) Logan confusion matrix



(b) Logan confusion Matrix normalized

Figure 2: YOLO confusion matrixes for Logan, Utah.

Table I

(a) Chosen Cities Based off Median Income and House Age

House Age	High Income	Middle Income	Low Income
Old Houses	Bountiful, Utah	Ogden, Utah	Price, Utah
New Houses	Saratoga Springs	St. George, Utah	(None)

to their relatively lower likelihood of appearing, to make sure the model would be trained on these less common building types. Despite this, building types such as schools and church buildings were still underrepresented and only appeared a hand full of times, due to their relatively low number even within larger buildings.

The model was run through hundreds of iterations until it was able to accurately and reliably predict building types in Logan. The model was then validated on images it had never seen to help prevent the model from “memorizing” the images, ensuring accurate results. The results of the validation of the model are shown in Figure 2.

C. Model Validation and Comparison

Once the model was trained and validated using the Logan, Utah data set, the model was then tested upon other cities in Utah to assess its viability in predicting building categories elsewhere in Utah. Using American Community Survey data, five other test cities were chosen based off the city’s median income level and the median age of the homes. These factors were theorized to have a significant impact on the architecture of the house and therefore the YOLO model’s ability to recognize and categorize them. The chosen cities can be seen in Table Ia.

Following the same method as previously described, building footprint data was collected and analyzed for each city. 200 aerial images of buildings were randomly chosen and annotated. These annotated data sets were then run through

the model using validation only. The results of these runs are given below in Figure 3. These results were then analysed to evaluate the ability of the model to predict in other cities, as well as assess if household income or house age were indicating factors.

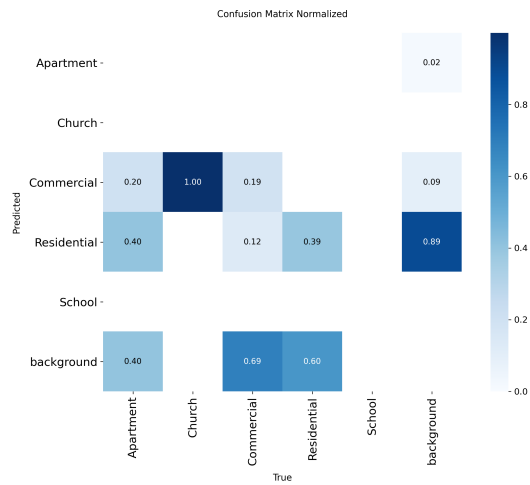
II. RESULTS AND DISCUSSION

A. Logan, Utah Results

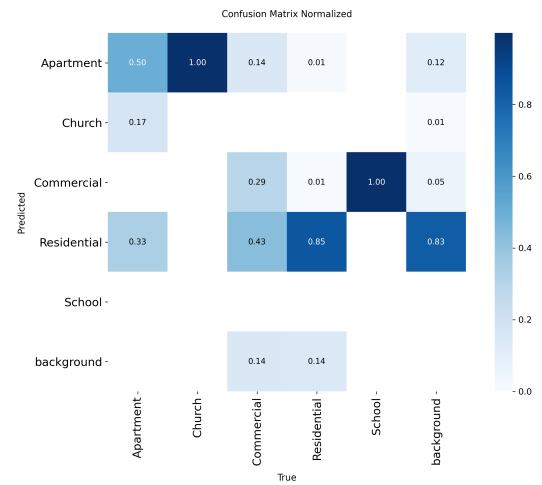
The results of the Logan trained YOLO model are shown in Figure 2. The desired output of the model was to have all the numbers in a diagonal line, which would indicate that the model is predicting the category of a building that is in line with the annotations made.

The model was able to accurately predict residential, commercial, and apartment building types almost 100% of the time. It did underperform with schools and churches. This is likely attributed to the fact that there were very low number of images of both building types provided during model training. Future training would seek out more images of these buildings or possibly find another way to identify them due to their relatively rare occurrence.

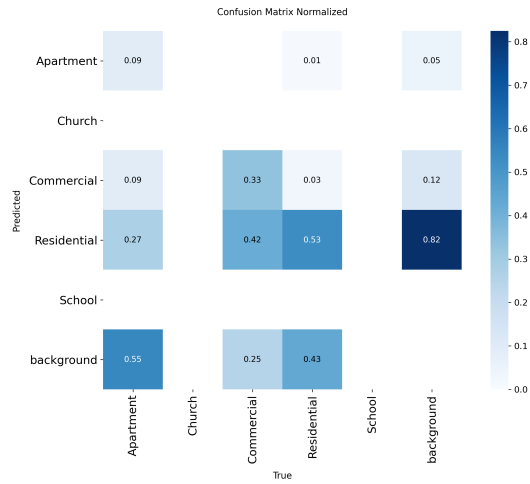
A important aspect to discuss is the issue with the model predicting different building types where there is no building. This is seen on the far right of each graph, where the background is given as the true value, but had predictions in every category. This error occurs from two sources. The first is due to the annotation. Auxiliary building such as backyard sheds were not consistently identified and annotated, but the model still picked them up and registered them. There were also times that only a small portion of a building was present in an image, and it was deemed to be too little of the building for the model to recognize, and thus was not annotated. However, the model in some cases still recognized and categorized the building at a low probability level, while in other cases it did not recognize partial buildings at all. The other source of error



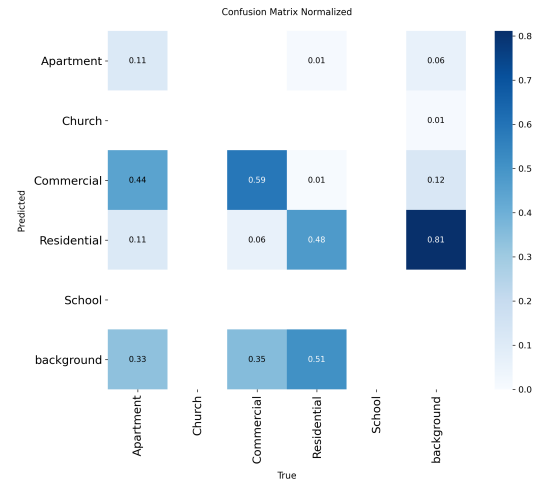
(a) Bountiful, Utah



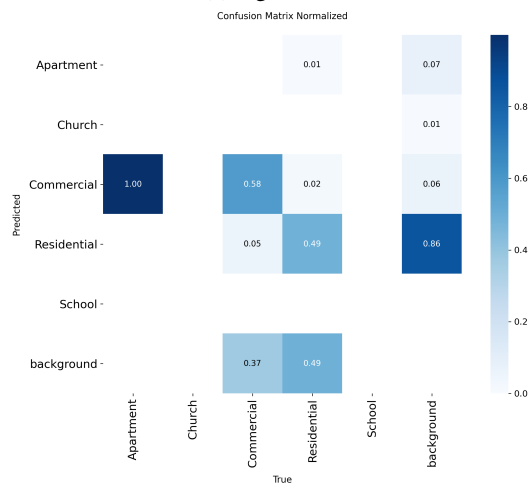
(b) Saratoga Springs, Utah



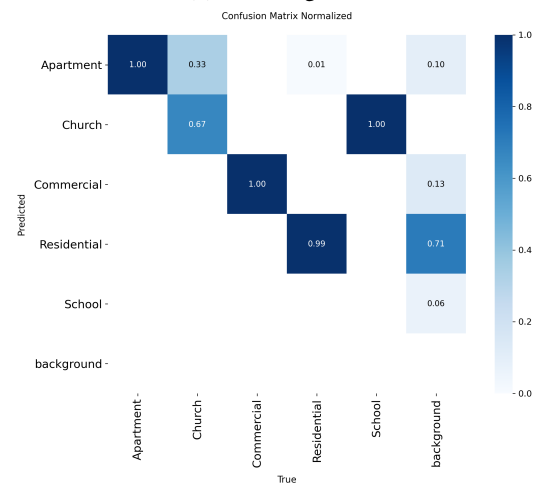
(c) Ogden, Utah



(d) St. George, Utah



(e) Price, Utah



(f) Logan Utah

Figure 3: Normalized YOLO confusion matrixes

was the quality of the photos used. The satellite images from Contextily were convenient and easy to access, but were not high quality. This resulted in the model sometimes interpreting things of solid colors, such as roads and covered parking, as a type of building. The second error could be solved using higher quality photos to help distinguish between these surfaces. The first error could be improved by implementing stricter guidelines on annotation to ensure consistency between data sets. However, in practice, the first error may be less relevant if only the primary building in each image matters to be identified.

B. Other Utah Cities Results

When the Logan model was applied to other cities in Utah, it produced mixed results, as seen in Figure 3. In all cities, the model struggled to correctly identify the different building types. Most of the building annotated and given to the model to identify were residential housing. Yet it misidentified residential building at least half of the time in most cities. Commercial buildings had even worse results, and practically all apartments, schools, and churches, were incorrectly identified. While it seemed to do better in Saratoga Springs with residential buildings, its accuracy with other building types is still far too low to be useful. It is not known why Saratoga Springs residential buildings were so well identified, with the city sharing little with Logan in respect to median income, house age, and location.

The model also struggled with simply recognizing buildings at all. Each city had a significant number of buildings that were predicated as background, meaning the model simply did not pick them up as buildings. When the building images were individually checked, some had trees or other shadows partially covering the roof which could help explain this behavior. Yet, many of the Logan building images have similar characteristics, so this is unlikely to be a factor. The image quality taken from Contextily could have played a role, yet images from both Logan and all other cities seemed to be of similar quality. So, it is uncertain whether higher quality images would have an effect on the model's accuracy in other cities.

While it was theorized that median income or house age would play a significant factor in the model's ability to recognize different building types, the results of the validations did not support this. The model performed roughly the same in every city it was not trained on. Neither house age or house median income seemed to make a difference in the model's ability to classify. While not tested, it also seems that a city's proximity is not a factor, due to all these cities being from Utah. It is possible, however, that cities outside of Utah would produce even worse results when validated with the Logan model.

Given the results of the validations, a process to develop a model for different locations would be more useful and viable than to create a universal model to be used in a region. It appears that building styles and types are unique enough between cities that a single unified model, at least

one mostly dependent on identifying building roofs, is not feasible. However, as will be discussed more in the future research section, adding other environmental variables to the classification, such as the presence of sidewalks, swing sets, etc. could allow for a more general model to be developed.

III. CONCLUSION AND FUTURE RESEARCH

This paper presented the results of using a deep learning model to classify building types from aerial imagery. Building footprints in Logan, Utah were used to identify buildings, images of individual buildings were used to train a YOLO model, and that model was then tested on predicated building categories from other cities in Utah in which it was not trained on.

Currently, this research is susceptible to many of the same problems as transportation models. Satellite and aerial imagery are not consistent around the world, having varying availability and image quality. It also may be outdated, especially in fast developing areas. The use of commercial drones could solve many of these problems, but they also come with their own problems. It would take some time to collect a whole city's worth of imagery, the flights would be dependent on the weather, and local and national regulations could present challenges. There are also potential ethical problems with such data collection, with drone collected imagery straddling the line between public and private information. This will need to be considered as further research is done into this area as to stay on the right side of the law [13].

This study is only in the beginning stages of development, with many potential future directions. As discussed previously, a model could be trained to detect other objects that could be used to develop socioeconomic data, such as cars, swing sets, patios, amount of foliage or green space around house, etc. Aerial imagery from drones could also be used to acquire higher quality imagery which could improve the training and accuracy of future models. This imagery could also be used to extract building footprints. The Microsoft Building footprint data set is expansive, but not exhaustive. It also features many notable errors in building footprint identification. Being able to extract more accurate footprints would help with clustering building correctly, which would in turn help with training future models.

Further research could also delve into how other remote sensing data could be utilized and incorporated into building classification. Data such as thermal radiation, nighttime light data, and vegetation index data. Such data could be used to predict statistics like income due to heat use, light use, or amount of vegetation on a property. Or it could give insights to the number of people living within a household, or whether children are present.

Proximity between buildings is another factor to consider. Similar building types are often next to each other, at least in the United States. Variables could be developed that look at the nearby buildings to the one in question and their predictions for classification. These new variables would influence the ultimate classification of the buildings. These factors could

run into issues in areas where mixed use housing is common, or where residential and commercial building are more intertwined.

All these improvements and variables could be used to form a larger model that would take all these variables into account when categorizing buildings. Using methods such as decision trees and random forests, this model would be able to create a process that would not only take YOLO's prediction of the building type into account, but also the other variables mentioned. This encompassing model would seek to not only categorize the buildings but also predict the type of household living in that building.

REFERENCES

- [1] Kamel Allaw et al. "Population estimation using geographic information system and remote sensing for unorganized areas". In: *Geoplanning: Journal of Geomatics and Planning* 7.2 (Nov. 1, 2020), pp. 75–86.
- [2] Casper Samsø Fibæk, Carsten Keßler, and Jamal Jokar Arsanjani. "A multi-sensor approach for characterising human-made structures by estimating area, volume and population based on sentinel data and deep learning". In: *International Journal of Applied Earth Observation and Geoinformation* 105 (Dec. 25, 2021), p. 102628.
- [3] Patrick Lehnert et al. "Proxying economic activity with daytime satellite imagery: filling data gaps across time and space". In: *PNAS Nexus* 2.4 (Apr. 1, 2023), pgad099.
- [4] Ye Zhang et al. "Landscape patterns and building functions for urban land-use classification from remote sensing images at the block level: A case study of wuchang district, wuhan, china". In: *Remote Sensing* 12.11 (Jan. 2020), p. 1831.
- [5] Shankar Acharya Kamarajugedda, Pradeep V Mandapaka, and Edmond Y.M Lo. "Assessing urban growth dynamics of major southeast asian cities using nighttime light data". In: *International Journal of Remote Sensing* 38.21 (Nov. 2, 2017), pp. 6073–6093.
- [6] Ying Chang et al. "A novel method of evaluating highway traffic prosperity based on nighttime light remote sensing". In: *Remote Sensing* 12.1 (Jan. 2020), p. 102.
- [7] Microsoft. *GlobalMLBuildingFootprints*. Version Python. Jan. 7, 2026. URL: <https://github.com/microsoft/GlobalMLBuildingFootprints> (visited on 01/07/2026).
- [8] Xinyue Ye et al. "Enhancing population data granularity: A comprehensive approach using lidar, POI, and quadratic programming". In: *Cities* 152 (Sept. 2024), p. 105223.
- [9] I. Colomina and P. Molina. "Unmanned aerial systems for photogrammetry and remote sensing: A review". In: *ISPRS Journal of Photogrammetry and Remote Sensing* 92 (June 2014), pp. 79–97. DOI: 10.1016/j.isprsjprs.2014.02.013.
- [10] Adam C. Watts, Vincent G. Ambrosia, and Everett A. Hinkley. "Unmanned Aircraft Systems in Remote Sensing and Scientific Research: Classification and Considerations of Use". In: *Remote Sensing* 4.6 (June 2012), pp. 1671–1692. DOI: 10.3390/rs4061671.
- [11] Bryce E. Berrett et al. "Large-Scale Reality Modeling of a University Campus Using Combined UAV and Terrestrial Photogrammetry for Historical Preservation and Practical Use". In: *Drones* 5.4 (Dec. 2021), p. 136. ISSN: 2504-446X. DOI: 10.3390/drones5040136. (Visited on 01/20/2026).
- [12] Gustavo Pereyra Irujo et al. "Open science drone toolkit: open source hardware and software for aerial data capture". In: *PLOS ONE* 18.4 (Apr. 6, 2023), e0284184.
- [13] Ola Hall and Ibrahim Wahab. "The use of drones in the spatial social sciences". In: *Drones* 5.4 (Dec. 2021), p. 112.
- [14] Glenn Jocher, Jing Qiu, and Ayush Chaurasia. *Ultralytics YOLO*. Version 8.0.0. Jan. 2023. URL: <https://github.com/ultralytics/ultralytics>.
- [15] Wahidya Nurkarim and Arie Wahyu Wijayanto. "Building footprint extraction and counting on very high-resolution satellite imagery using object detection deep learning framework". In: *Earth Science Informatics* 16.1 (Mar. 1, 2023), pp. 515–532.
- [16] Mayank Sharma and Rahul Dev Garg. "Building footprint extraction from aerial photogrammetric point cloud data using its geometric features". In: *Journal of Building Engineering* 76 (Oct. 1, 2023), p. 107387.
- [17] Dani Arribas-Bel. *contextily: context geo tiles in Python*. 2023. URL: <https://contextily.readthedocs.io>.
- [18] Maxim Tkachenko et al. *Label Studio: Data labeling software*. Open source software available from <https://github.com/HumanSignal/label-studio>. 2020–2025. URL: <https://github.com/HumanSignal/label-studio>.